Literature Review: Sports Mitra

1. Introduction

This review is for a research project on StatsBomb soccer event data where each event is encoded as an additive sum of feature-group embeddings (player/team identity, spatial/temporal context, game state, etc.). The task is next-event prediction with three transformer-based models: an identity-only model, a situational-only model, and a full model combining both. The core scientific questions are:

- How much of the next-event prediction signal is carried by identity vs. situational features?
- How does a transformer internally combine these additive summands — do different heads/layers specialize, and can we causally trace identity vs. situation circuits?

The review is organized into:

- Section 2 — Personal reading notes on the four most central papers.
- Section 3 (A) — Soccer event modeling and representation.
- Section 4 (B) — Mechanistic interpretability and transformer circuits.
- Section 5 (D) — Full reference list.
- Appendix — Gaps and further reading.

2. Personal Reading Notes

2.1 Seq2Event (Simpson et al., 2022)

A decent baseline for prediction performance: a next-event predictor with a softmax output, evaluated across multiple architectures (Elman-RNN, LSTM, GRU, Transformer). The Transformer was ultimately used for downstream analysis because of its parallel-processing speed advantage — notably, an LSTM variant actually achieved the best raw test loss (1.4h vs. 15.5h training time was the deciding factor, not accuracy). Experiment 2 in the paper (spatial analysis of shot-prediction probability) is directly replicable to test context-awareness in our own models.

Limitations: only four collapsed event types (pass, dribble, cross, shot); concatenates the action and continuous-feature embeddings rather than summing them, which departs from the additive structure we need for interpretability; trained on a comparatively small dataset (138 matches) relative to the scale of StatsBomb data available here.

2.2 RisingBALLER (Adjileye, 2024)

Introduces spatial and temporal positional embeddings alongside player and team identity embeddings, combined via elementwise summation — the closest existing precedent for our additive embedding design. Uses masked player prediction (players as tokens, matches as sentences) as a pretraining task.

On the team/player identity question: with the team affiliation embedding included, player embeddings still cluster primarily by team, and only shift toward capturing role/style similarity once the team embedding is removed. This is better described as an emergent, empirically-observed dominance of team identity over player-specific signal(this is actually opposite of what other papers found and could be due to small dataset size good to be aware of), rather than a designed "hierarchy" — the four embedding components (player ID, position, team, temporal stats) are architecturally parallel and summed with no built-in priority ordering.

Limitations: small dataset (1,792 matches, single season), single GPU, single transformer layer — the authors themselves flag that team-identity dominance may partly be a small-data artifact rather than a general property.

2.3 Romero et al. (2025/2026)

A similar concept to the identity-vs-context question above, but using an MLP-based variant of the LEM framework on the Wyscout dataset rather than a transformer on StatsBomb. Reports a result that appears to conflict with RisingBALLER: in the absence of an explicit team embedding, the model encodes team-specific style directly into the player embedding (player embeddings cluster by team when team information is not separately supplied) — and this clustering effect disappears once an explicit team embedding is included.

Flag: a genuine point of tension between RisingBALLER and Romero et al.

RisingBALLER: WITH the team embedding included, player embeddings still cluster by team (removing it is what causes clustering to shift toward role/style).

Romero et al.: WITH the team embedding included, team-based clustering in player embeddings disappears (shifts toward style); it is the ABSENCE of the team embedding that causes player embeddings to compensate by encoding team identity.

These are opposite directions for the same manipulation. Plausible explanations include architectural differences (transformer-over-player-tokens vs. MLP-over-event-features) or dataset differences (StatsBomb 2015/16 top-5 leagues vs. Wyscout) — worth investigating directly in our own setup rather than assuming either result generalizes.

> Update: dug into this more. The likely explanation isn't that the two papers found contradictory facts — it's that the architectures put the team embedding in different places. RisingBALLER sums the team embedding into the *same* per-token vector as the player embedding, so both flow through the same residual stream and end up sharing structure — team embeddings there seem to reinforce team clustering in player embeddings. Romero's model is an MLP where the team embedding is a separate input slot, so the network can dump team-specific info into that slot instead of the player embedding — team embedding there seems to relieve player embeddings of carrying team info. So this might be an architecture-dependent effect (token-transformer-with-summed-embeddings vs. MLP-with-separate-slots) rather than a real disagreement about the underlying phenomenon. No paper has directly tested this side by side — this is a real gap and something we could test ourselves by training both regimes with/without the team embedding.

Limitations: MLP and XGBoost only, no transformer-based ablation and no mechanistic interpretability — a strong empirical precedent for the exclusive-model comparison design, but it does not extend to circuit-level analysis.

2.4 NMSTPP (Yeung, Sit & Fujii)

If the project scope expands from single next-event-type prediction to full spatio-temporal point process prediction (jointly modeling inter-event time, spatial zone, and action type), this is the reference architecture to build on.

Reference: arXiv:2302.09276; published as Yeung, C., Sit, T., & Fujii, K., "Transformer-based neural marked spatio temporal point process model for analyzing football match events," Applied Intelligence 55, 335 (2025), https://doi.org/10.1007/s10489-024-05996-9.

2.5 EventGPT (Hong et al., 2025)

A GPT-style model that predicts the next event (type, location, timing, outcome) conditioned on player identity, trained on five seasons of Premier League data. Everything — player identity included — goes into one shared token vocabulary, so there's no explicit additive split of identity vs. situational groups like we're doing. The useful part: they run counterfactual player-ID substitution (swap one player's ID token for another and see how the predicted outcome changes) to measure player impact. That's a usable technique for us even without their architecture — we could swap identity embeddings the same way and measure the shift in next-event predictions.

Limitations: no additive embedding structure, so it doesn't help with the mechanistic side directly; useful mainly for the counterfactual-substitution idea.

3. Section A: Soccer Event Modeling & Representation

3.1 Event Tokenization and Vocabulary Schemes

Standardized action schemas are crucial because they determine how you split features into "identity" vs. "situational" groups and how many parameters each group gets.

SPADL / VAEP

Decroos et al. introduce SPADL (Soccer Player Action Description Language) as a unified representation of on-ball actions, decomposing each event into fields such as action type, start and end coordinates, body part, result, and player/team identifiers [3]. On top of this representation, VAEP trains models that estimate the change in scoring/conceding probabilities attributable to each action [3].

For our project: SPADL offers a canonical feature taxonomy — identity features: player ID, team ID; situational features: action type, location, body part, result, time, match context. It is widely adopted, ensuring compatibility with Seq2Event and other models.

Seq2Event and Tokenized Event Sequences

Simpson et al. treat a match as a sequence of event tokens, each token encoding SPADL-like fields plus context, and train RNN/transformer models to predict the next event's type, location, and time [1]. Their architecture uses standard learned embeddings for each categorical feature, concatenated (not summed) into a single input vector per time step [1].

Implications: direct precedent for next-event prediction on StatsBomb-style data with transformer architectures; validates the "events as tokens" perspective; suggests evaluation metrics such as accuracy/cross-entropy over next event type, binned location, and inter-event time.

SoccerTransformer (Rovshitz & Puzis, 2025)

Another events-as-tokens transformer, pretrained self-supervised on event sequences, then fine-tuned for tasks like attack-outcome prediction and player rating. Keeps player/team IDs as ordinary categorical features and attends across features within each event plus across the sequence, but doesn't separate identity from situational features additively. Relevant mainly as a second precedent (besides Seq2Event) for the token-sequence approach on soccer data [12].

Foundation Models for Soccer Event Sequences

Two recent works generalize Seq2Event's idea to larger-scale setups. Baron, Hocevar & Salehe propose "A Foundation Model for Soccer," using a transformer decoder to predict subsequent actions from sequences of SPADL-like actions, trained on three seasons of professional match data and compared against Markov and MLP baselines [19]. Mendes-Neves et al. introduce the

Large Events Model (LEM), a large-scale event-level transformer pre-trained on multiple seasons of events and then adapted for tasks like outcome forecasting and action valuation [2].

For our purposes: these papers justify pretraining on raw event sequences before specializing to next-event prediction, and provide data-scale and training-regime references. Their embeddings are standard learned categorical embeddings, structurally aligned with (though not identical to) our additive design.

3.2 Additive / Multi-Embedding Architectures

Our setup explicitly sums multiple feature-group embeddings into each event vector, so work that does this (or something close) is particularly relevant.

RisingBALLER: Player Token with Decomposed Embeddings

RisingBALLER models players as tokens and matches as "sentences," focusing on player representation rather than event sequences per se [5]. Each player token is represented as the elementwise sum of a unique player ID embedding, positional embedding, team embedding, and temporal information embedding [5], processed by a transformer encoder.

Relevance: the closest direct precedent for an additive multi-embedding design; shows that (ID + spatial + team + time) additive decomposition is practical and supports downstream analytics; offers a blueprint for separating identity (player ID, team) vs. context (position, time) in embedding space while still using a standard transformer.

TabTransformer-Based Universal Event Representation

Yang et al. use TabTransformer to generate a universal dense event embedding from StatsBomb's high-cardinality categorical variables (including player ID and team) [6]. They first learn feature-level self-attention purely within the tabular space, then export a single dense vector per event, with an explicit design goal of reducing player/team identity bias in the learned representation [6].

Concretely: 28 categorical fields and 15 continuous fields go in, player/team IDs and other leakage-prone fields are stripped out, and the output is a 911-dim dense event vector used downstream for xG, VAEP-style value estimates, and player-style similarity search.

Relevance: a mirror image of our project — they strip identity information, whereas we isolate and analyze it. Their feature grouping provides a concrete list of which categorical features to treat as identity vs. situational.

Player-Team Heterogeneous Interaction Graph Transformer (Wang et al., 2025)

Builds a graph where players and teams are separate node types, and runs a graph-augmented transformer on Wyscout data for match outcome prediction, fusing player-level and team-level representations in a "Match Comparison Transformer." Player and team representations stay as

separate objects rather than being summed into one vector, and there's no explicit identity-vs-situational analysis. Relevant as a different way of keeping identity and context separate (graph structure instead of additive embeddings) — worth a mention as an alternative design choice, but doesn't directly extend our approach [20].

NMSTPP: Neural Marked Spatio-Temporal Point Process

Yeung et al. model football events as a neural marked spatio-temporal point process: a transformer encodes history, and outputs parameterize the intensity for the next event's time, space, and type [4]. Discrete event types are treated as marks and continuous coordinates/time as continuous marks, with an additive embedding scheme over categorical attributes and discretized spatial bins [4].

Relevance: a principled way to model the joint distribution over time, location, and type; separate output heads for time vs. coordinates vs. type can be adopted alongside our additive input structure.

3.3 Identity vs. Situational Feature Separation

Our project's central design choice is to separate identity-based from situational features and train exclusive transformers on each. There is one paper that studies nearly the same question at the behavioral level.

Romero et al.: Impact of Team and Player Information

Romero et al. analyze next-event prediction in soccer under different information sets, using MLP-based models with or without team/player information, benchmarked against heuristic baselines and XGBoost [7]. They quantify the performance gain from adding team and player IDs on top of context-only features, across different event types and match situations [7].

Relevance: the closest direct conceptual precedent to our identity-only vs. situational-only vs. full setup. Their experimental design — training model variants with inclusion/exclusion of identity features and comparing performance — is an excellent template for our exclusive vs. full model comparison. See Section 2.3 above for the point of tension between this paper's findings and RisingBALLER's regarding what happens to player embeddings when team identity is included vs. excluded.

Most papers just compare individual vs team features not identity vs situational so this is a real gap.

3.4 Objectives and Evaluation in Event-Sequence Models

Across Seq2Event, NMSTPP, LEM, and EventGPT, a fairly consistent set of objectives and evaluation metrics emerges:

- Autoregressive next-event prediction: cross-entropy over next event type, location bins, or parameterized distributions for continuous targets [1][4].
- Log-likelihood / negative log-likelihood, especially in NMSTPP's point process framework [4].
- Downstream metrics: attack/possession outcome prediction, action value estimation (VAEP-style [3], or LEM [2] / EventGPT [8]).
- Identity-aware evaluations: performance broken down by player or team [1][7]; performance change when swapping or withholding identity features [7][8].

For our study, this suggests using per-event-type and per-phase breakdowns of predictive performance to see where identity-only and situational-only models diverge, and considering simple downstream tasks (e.g., "does this action increase the chance of a goal in the next K actions") as additional targets for comparing identity vs. situational contributions in value terms.

4. Section B: Mechanistic Interpretability & Transformer Circuits

4.1 Foundational Circuit Theory

Anthropic's Transformer Circuits Framework

Elhage et al. provide a mathematical framework for viewing transformers as compositions of linear operations plus simple nonlinearities, with emphasis on the residual stream and QK/OV decomposition of attention [9]. The residual stream at each layer is a sum of contributions from embeddings, attention heads, and MLPs; direct logit attribution decomposes output logits as the sum of contributions from each layer or head — a decomposition that is linear by construction (aside from normalization effects), aligning with our additive embedding hypothesis.

Relevance: supplies the language and formulas needed to discuss the contribution of each embedding summand (identity vs. situational) as it propagates through the network; underlies both DePass-style decomposed forward passes and IOI-style circuit tracing.

IOI Circuit as Worked Example (IOI - indirect object identification)

Wang et al. reverse-engineer a circuit in GPT-2 small responsible for indirect object identification [10], deriving a methodology for causal tracing and activation patching (substituting "clean" representations into a "corrupted" run to localize responsible components) and actually demonstrating circuit discovery via functionally-grouped attention heads.

Relevance: although IOI is a language task, the methodology is relevant for us— we can treat a specific event pattern as our IOI-like behavior and follow similar steps to identify responsible heads/layers.

4.2 Causal Tracing and Patching Methodology

Time-Series Adaptation of IOI

Kalnāre et al. adapt IOI-style MI tools to transformer-based time-series classifiers [11], using activation patching on non-token, numeric sequences and training sparse autoencoders (SAEs) on internal representations, causally verified via patching and ablation.

Relevance: a direct blueprint for applying MI techniques to non-text event sequences, and a practical reference for deciding which layers to patch, how to sample inputs, and how to interpret SAEs in a non-linguistic domain.

Path Patching

Goldowsky-Dill et al. introduce path patching — instead of ablating a whole component, you patch activations only along a specific *path* (a chain of connections between components, e.g. head→MLP→layer) using a counterfactual input, holding everything else fixed, then measure the effect on the output [13]. It's a way to test a specific hypothesis like "this behavior lives only in these three connections" rather than doing coarse whole-component ablation.

Relevance: this is a good fit for testing whether identity and situational information travel through separate paths in our model. We could define a path hypothesis like "situational features drive shot-vs-pass decisions late in the game; identity paths don't matter here," then patch identity-carrying paths with neutral values while holding situational paths fixed, and see if predictions change. Gives a sharper test than just zeroing out the whole identity embedding.

4.3 Additive / Decomposed Forward-Pass Methods

DePass: Decomposed Forward Pass for Transformers

Hong et al. introduce DePass, running a single decomposed forward pass while tracking how user-defined groups of input features contribute to every hidden state and output, achieving lossless additive decomposition through all layers [14].

Relevance: tailor-made for our project — our event embedding is an explicit sum of known groups (identity vs. situational), and DePass allows that decomposition to be carried through the entire model, offering a more principled and efficient way to quantify each group's contribution than naive ablation or gradient-based methods.

Interpretable Additive Tabular Transformer Networks (NATT)

Thielmann et al. combine neural additive models with transformers for tabular data — each feature (or feature group) gets its own additive contribution to the output, but a transformer computes that contribution so it can still capture nonlinear relationships [15]. So the final prediction is literally a sum of per-feature-group terms, and each term can be read off directly instead of needing a post-hoc explainer like SHAP.

Relevance: this is basically the interpretability version of what we're doing at the embedding level. We could put a NATT-style head on top of our event transformer, treating identity and situational groups as the "features," and get a direct additive breakdown of how much each group contributes to the output — then compare that global additive picture against what path patching finds at the circuit level. If they agree (situational-heavy layers = situational-heavy NATT contribution), that's a nice cross-check.

4.4 Limitations and Failure Modes of Additive/Linear Explanations

Attention Mechanisms Don't Learn Additive Models

Leemann et al. show that standard transformer attention is structurally incompatible with additive models often used in feature-importance explanations, and that naive additive attributions can substantially mischaracterize feature interactions [16].

Relevance: a crucial caveat — even if identity vs. situational contributions can be linearly decomposed in the residual stream, the underlying attention mechanism may be highly non-additive. We should treat linear decomposition as an approximate interpretive tool, not a literal mechanistic description, unless further evidence is provided by causal experiments.

Self-Repair and Backup Circuits

Gong et al. demonstrate that ablating a component in transformer circuits can trigger self-repair, where dormant backup circuits partially restore performance, and introduce Conditional Co-Ablation (CoAx) to quantify how much each remaining unit's ablation effect grows once a primary set has been removed [17].

Relevance: explains why ablation experiments in the full model can under-report a feature group's true importance; strongly supports training exclusive identity-only and situational-only models and interpreting full-model ablations only in light of those.

Does Superposition Even Apply Here?

Yıldırım tests whether transformers trained on time-series forecasting rely on superposition (packing many features into overlapping directions) the way language models do [18]. Using sparse dictionaries at different sizes, they find representations stay sparse, performance barely changes even with a 4x bigger dictionary (~0.2% difference), and directly intervening on the most "active" features barely moves the forecast. Conclusion: strong superposition isn't necessary for these models to do well.

Relevance: this is a caution for us in the other direction from Leemann et al. above. It's tempting to assume our transformer will naturally organize identity and situational signal into separate, findable directions in the residual stream. Yıldırım's result suggests time-series/event transformers might not need rich compositional structure at all to perform well — so there's no guarantee our model will factor things into clean "identity" vs. "situational" directions unless our additive embedding design is actually forcing that structure (which it might be, since we're explicitly summing separate groups going in — this is a genuine architectural difference from what Yıldırım studied, so it's not a direct contradiction, just a reason not to assume disentanglement for free).

5. Section D: Full Reference List

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Appendix: Gaps and Recommended Next Reading

Is anyone doing the exact three-model comparison + circuit analysis?

Based on the surveyed literature, no existing work combines (1) a three-model setup (identity-only, situational-only, full), and (2) a systematic mechanistic interpretability analysis (DePass, EAP, path patching, CoAx) to trace identity vs. situational pathways inside the full model.

Romero et al. come closest on the modeling side but stop at performance comparisons without transformer-circuit analysis [7]. DePass provides the decomposition machinery but has not yet been applied to soccer or similar additive event embeddings [14].

Update: broader search confirms this gap still stands as of the 2025–2026 literature — nothing combines the three-model comparison with systematic circuit-level tracing. Also worth flagging: no paper has directly tested why RisingBALLER and Romero et al. find opposite effects of team embeddings on player-embedding clustering (see Section 2.3) — this looks like an open question we could actually answer ourselves by training both a token-transformer and an MLP version with/without team embeddings and comparing.

Checklist for our model

Team vs. player dependence — check whether, in your model, team identity dominates player identity in the embeddings, or the reverse (RisingBALLER found team dominance; Romero found the opposite depending on setup).

Does the RisingBALLER-vs-Romero effect depend on architecture — test whether team-embedding-in-same-vector (token-transformer style) vs. team-embedding-as-separate-slot (MLP style) actually causes the clustering difference in your own setup, since nobody's tested this side by side.

Does your model need superposition — check whether your identity and situational features are packed into overlapping/shared directions, or whether they stay in separable subspaces (Yıldırım found time-series transformers often don't need superposition to perform well — don't assume yours does either).

Is the additive decomposition actually linear inside the model — verify whether the residual stream sum of identity + situational embeddings stays roughly separable through the layers, or whether attention mixes them non-additively (Leemann et al.'s warning that attention isn't naturally additive).

Self-repair / backup circuits — check whether ablating the identity or situational pathway in your full model gets partially compensated by other components (Gong et al.'s finding), which would mean single-ablation experiments understate a feature group's true importance — this is why you're training exclusive identity-only/situational-only models too.

Small-dataset artifact check — check whether your identity-vs-situational findings hold up as you scale up training data, since RisingBALLER's authors flagged that their team-dominance result might just be a small-dataset artifact (1,792 matches, single season).

Concatenation vs. summation sensitivity — check whether your results would look different if you concatenated identity/situational embeddings instead of summing them (Seq2Event used concatenation; you're deliberately using summation for interpretability — worth confirming this choice actually matters empirically).

Where in the network identity vs. situational information gets used — use path patching to test whether specific behaviors (e.g. shot vs. pass decisions) route through identity-heavy paths, situational-heavy paths, or both, rather than assuming it from ablations alone.

Additive attribution vs. circuit-level attribution agreement — check whether a NATT-style global additive breakdown of identity vs. situational contribution agrees with what path patching finds at the circuit level; disagreement here would be itself an interesting finding.

Identity leakage into "situational" model — check whether your situational-only model secretly recovers identity-like information indirectly (e.g., through play style or positioning patterns), the way Yang et al. worried about "ability confounding" leaking into supposedly identity-free representations.

Consistency across event types and match phases — check whether identity vs. situational dominance shifts depending on event type or game phase, rather than being a fixed global property of the model (hinted at in Section 3.4's evaluation objectives).